

Dear Editors,

Please consider our manuscript, “Charge-self-consistent equivariant potentials for energetic molecular crystals: static gains and nonequilibrium limits,” for publication as an Article in *npj Computational Materials*.

This work addresses a materials-modeling question that fits the journal well: when does explicit self-consistent electrostatic response improve machine-learned interatomic potentials for molecular crystals, and where does that improvement fail to transfer to nonequilibrium validation? We focus on CL-20/HMX as the motivating energetic-material target and develop CSC-MACE, a charge-self-consistent extension of MACE in which the same equivariant atomic features parameterize both the short-range energy and an environment-dependent charge-equilibration contribution to energies, forces, and stresses within one periodic total-energy model.

The manuscript makes three contributions. First, it formulates the problem in terms of a concrete physical limitation of finite-cutoff local models under strong compression, where intermolecular rearrangement, polarization, and charge redistribution evolve together. Second, it specifies a variational charge-aware extension of an equivariant interatomic potential in which explicit self-consistency is isolated against both a short-range baseline and a simpler static-charge reference, and the electrostatic branch is shown to be measurably active rather than numerically inert. Third, it evaluates this formulation across a structured evidence ladder comprising a public reference system (OpenREACT), two public surrogate systems (Transition1x and SPICE), and a final CL-20/HMX benchmark.

The final target benchmark is central to the paper. On the real CL-20/HMX static multiseed benchmark, the self-consistent branch achieves the strongest energy ranking, followed by the static-charge approximation, while force MAE remains effectively tied across the three families. That ordering now survives not only the original contiguous slice, the cleaner material-balanced split-v2 rerun, and the order-recovered block-holdout split-v3 rerun, but also a stronger temperature-gap split-v4 rerun. The associated QEq diagnostics show that the electrostatic branches remain active and well behaved on the target rather than collapsing to a numerically irrelevant correction. The nonequilibrium results reveal the boundary of the method rather than a simple success story. Transition1x produces family-level separation in a demanding regime, yet the separation favors the short-range baseline after comparable early bond-network rearrangement. SPICE instead shows frame-selective synchronized collapse with weak family separation. On real CL-20/HMX rollouts, the current evidence still points mainly to frame-dependent degradation rather than a reproducible family-level hierarchy, even when the observables are upgraded from drift to molecular integrity, key-bond survival, a QEq-versus-failure bridge analysis, a dedicated multiseed nitro-focus rerun, a threshold-aware fragile-frame summary, and stronger adaptive and train-matched self-consistent ablations on the most fragile target frames. The newest dense fragile-window follow-up sharpens the positive side of that story without overturning the broader boundary: train-matched self-consistency does not consistently delay the earliest short-bond onset, but it does improve endpoint structural survival on a narrow frame-64–72 window, with the clearest local gain on frame 72 in final molecular integrity, delayed first-integrity crossing, and a small nitro-survival increment. We believe this is useful for the computational materials community because it shows both where explicit self-consistency helps and where that help still fails to transfer, and it does so through a chemically demanding benchmark rather than only through public surrogate datasets.

The manuscript should be of interest to readers working at the interface of machine-learned interatomic potentials, atomistic simulation of materials under extreme conditions, energetic molecular

crystals, and physically informed materials modeling. It aligns with *npj Computational Materials* because the central contribution is not a generic machine-learning benchmark, but a materials-motivated analysis of how charge-aware equivariant models change static ranking, bond survival, fragment integrity, and trajectory failure modes across both public benchmarks and a final energetic-material target. The paper is also explicit about the present target-data boundary: the current CL-20/HMX release supports a real force-labeled energetic-material comparison, and the underlying data payload can now be traced to two named upstream DeepMD-format packages recovered in the local workspace snapshot. Within the author team, including Dongping Chen as both a coauthor of the present manuscript and an author on the PCCP 2024 CL-20/HMX paper, the staged subset is confirmed to follow that published CL-20/HMX sampling lineage, with retained scale factors 0.96, 1.00, and 1.04 and retained temperatures 300, 500, 1000, 2000, 3000, and 4000 K, and with retained systems that can be matched back to specific temperature/scale conditions by frame order after conversion. The final subset was not relabeled after receipt; instead, structures with any cell edge below 1 nm (10 Å) were filtered out under the collaboration’s benchmark-quality rule that periodic structures with any cell edge below 10 Å should not enter the final benchmark. Basic labeling settings are also now known within the collaboration: CP2K with PBE, GTH pseudopotentials, DZVP-MOLOPT basis functions, Grimme D3 dispersion, and auxiliary plane-wave cutoffs of 60 Ry and 400 Ry. What remains incomplete is not the existence of the data arrays themselves, but the reviewer-complete reproducibility record that would unambiguously specify the upstream per-frame thermodynamic metadata, trajectory-history metadata, and shock-generation workflow. We therefore treat that remaining provenance limit as part of the result rather than as hidden missing information.

This manuscript is original, has not been published previously, and is not under consideration elsewhere.

Sincerely,

Xiaohua Liu

State Key Laboratory of Precision Blasting, Jiangnan University

Corresponding author: xiaohualiu@jhun.edu.cn